A Spatial–Spectral Collaborative Method for Radiometric Cross-Calibration of Ocean Color Satellite Sensors

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***Abstract*—Radiometric calibration represents an indispensable foundation for quantitative ocean color remote sensing, with cross-calibration serving as a cornerstone methodology. However, existing cross-calibration methods primarily rely on linear calibration models. These approaches struggle to accurately characterize nonlinear response residuals across complex environments and overlook both the spatial correlations among adjacent pixels and the spectral correlations between distinct bands. To address these challenges, this study departs from the conventional coefficient-fitting paradigm and proposes an innovative feature-driven radiometric cross-calibration method. By learning high-dimensional image features, this approach establishes a direct sensor-to-sensor mapping, obviating the need for explicit calibration coefficients. In particular, designed for the 3-D spatial–spectral characteristics of satellite imagery, the proposed dual-branch spatial–spectral collaborative network (DB-SSCN) synergizes local spatial features and global spectral dependencies, incorporating a band-weighted loss function to balance contributions across different bands. In a case study utilizing the Chinese Ocean Color and Temperature Scanner (COCTS) onboard the Haiyang-1-D (HY-1D) satellite, the calibration results exhibit strong consistency with the standard radiance products of Aqua MODIS, yielding a mean squared error (mse) of 0.2661 and a mean relative error (MRE) of 1.18%. Furthermore, the developed algorithm demonstrates exceptional spatial and spectral preservation capabilities, achieving a structural similarity index measure (SSIM) of 0.9879 and a spectral angle (SA) of 0.28°. The method effectively suppresses striping artifacts, thereby significantly improving the quality of the radiometric calibration products. Application experiments confirm that the method exhibits robust applicability across diverse water environments.**

***Index Terms*—Deep learning, Haiyang-1-D (HY-1D) Chinese Ocean Color and Temperature Scanner (COCTS), Ocean Color Satellite, radiometric cross-calibration, spatial–spectral collaborative network.**

I. Introduction

OCEAN color satellites, leveraging their advantages of wide swath and periodic observation, have become the primary approach for quantitative monitoring of the marine environment [1], [2]. Radiometric calibration is crucial for the quantitative applications of ocean color satellites, thereby directly determining the reliability of derived remote sensing products. On-orbit radiometric calibration methods for ocean color satellite sensors primarily encompass on-board calibration [3], [4], vicarious calibration [5], [6], and cross-calibration [7], [8]. On-board calibration offers high precision and is insensitive to variations in Earth's surface environment and atmospheric conditions. However, this approach necessitates additional calibration devices. Furthermore, as these instruments degrade over time, their calibration accuracy progressively deteriorates [9]. On the contrary, vicarious calibration imposes stringent requirements on the calibration site and observation conditions, relying heavily on high-precision in situ measurements [10].

Cross-calibration effectively overcomes the spatiotemporal limitations of conventional calibration methods and serves as a critical complement to on-board and vicarious calibration. Researchers have successfully achieved precise radiometric calibration across multiple satellite instruments using cross-calibration techniques. For instance, using simultaneous nadir overpass (SNO) data from Terra MODIS and the Advanced Very High Resolution Radiometer (AVHRR) aboard NOAA-16, Heidinger et al. [11] developed a regression model to relate AVHRR digital number (DN) values to MODIS reflectance. Similarly, Liu et al. [12] performed cross-calibration for the multichannel visible infrared scanning radiometer onboard the Fengyun-1-D satellite, using Terra

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MODIS as reference data. Cao et al. [7] collected nearly concurrent data over radiometrically stable targets, such as deserts and ocean gyres, to facilitate cross-calibration between the Visible/Infrared Imager Radiometer Suite (VIIRS) and MODIS. Furthermore, Angal et al. [13] conducted cross-calibration between the Sentinel-2 MultiSpectral Instrument and MODIS using desert regions as pseudoinvariant calibration sites. Although the aforementioned methods incorporate multiple factors, their final formulations converge to deterministic calibration coefficients (gain and bias), thereby constraining them to a linear regression paradigm. When addressing complex scenarios such as coastal waters, this paradigm struggles to characterize the nonlinear response residuals of sensors across varying dynamic ranges [14]. Furthermore, it fails to effectively utilize the latent spatial correlations among pixels and the spectral correlations between bands.

In recent years, deep-learning methods, distinguished by their robust feature extraction capabilities, have been widely adopted across various remote sensing domains, including satellite image processing [15], [16], [17], disaster monitoring [18], [19], [20], and object detection [21], [22]. However, their application in radiometric cross-calibration remains limited. Existing studies have predominantly focused on employing deep neural networks to estimate linear calibration coefficients, effectively replicating conventional linear calibration models. For example, Chen et al. [23] used a multilayer perceptron (MLP) to model the relationship between sensor status parameters and calibration coefficients. Li et al. [24] and Tang et al. [25] proposed convolutional neural network-based approaches for radiometric calibration to address sensor degradation over time. Peng et al. [26] developed a radiometric cross-calibration method that integrates a radiative transfer model with a deep neural network, facilitating precise mapping between the calibration reference and target spectral bands. These methods persist in the conventional paradigm of employing neural networks merely to estimate calibration coefficients. Consequently, they remain fundamentally constrained within the traditional linear calibration framework.

To address the aforementioned limitations, this study proposes a novel deep-learning-based cross-calibration method. Departing from the conventional practice of solving for explicit calibration coefficients, our approach leverages high-dimensional image features to establish a direct, end-to-end mapping from DN to top-of-atmosphere (TOA) radiance. The primary contributions of this work are summarized as follows.

1) *Pioneering a Novel Cross-Calibration Paradigm:* Shifting away from conventional methods relying on linear calibration coefficients, this study establishes a direct, end-to-end mapping from DN to radiance, bypassing the dependence on explicit coefficient inversion.
2) *Designing a Dual-Branch Spatial–Spectral Collaborative Network:* This architecture synergizes local spatial refinement with global spectral dependencies, thereby surmounting the limitations of conventional pixelwise methods.

TABLE I
SENSOR SPECIFICATIONS

Sensor	Band Number	Spectral Range (nm)	GSD (m)	Swath (km)
COCTS	1	402-442	1100	2900
	2	433-453		
	3	480-500		
	4	510-530		
	5	555-575		
	6	660-680		
	7	730-770		
	8	845-885		
MODIS	4	545-565	1000	2330
	8	405-420		
	9	438-448		
	10	483-493		
	11	526-536		
	13	662-672		
	15	743-753		
	16	862-877		
VIIRS	1	402-442	750	3000
	2	436-454		
	3	478-498		
	5	662-682		
	7	846-885		

II. DATA AND METHODS

A. Satellite Data

To train and validate the proposed cross-calibration model, the Chinese Ocean Color and Temperature Scanner (COCTS) onboard the Haiyang-1-D (HY-1D) satellite was selected as the target sensor in this study. Launched in June 2020, HY-1D is a new-generation Chinese medium-to-high-resolution oceanic optical satellite. The onboard COCTS sensor is configured with ten bands (Table I), comprising eight visible and near-infrared (VNIR) bands (Bands 1–8) and two thermal infrared (TIR) bands (Bands 9 and 10). It features a wide swath of 2900 km and a spatial resolution of 1100 m, enabling the large-scale and precise monitoring of environmental parameters such as ocean color and sea surface temperature. The COCTS satellite data utilized in this study were acquired from the Chinese Ocean Satellite Data Service System (https://osdds.nsoas.org.cn).

In the selection of reference sensors, MODIS onboard the Aqua satellite and VIIRS onboard the Suomi-NPP (SNPP) satellite are regarded as ideal benchmarks for cross-calibration, owing to their comprehensive on-board calibration systems and the advantage of daily global coverage [13], [27]. Both platforms operate in coordinated afternoon orbits with equatorial crossing times of approximately 13:30, ensuring excellent temporal synchronization with HY-1D COCTS. This alignment effectively mitigates environmental errors induced by discrepancies in observation times and diurnal variations in solar illumination. Although both are high-quality sources, Aqua MODIS was selected as the primary reference for model training due to its superior spectral response matching with COCTS in the VNIR bands (Table I). Meanwhile, SNPP VIIRS was introduced as an independent source to validate

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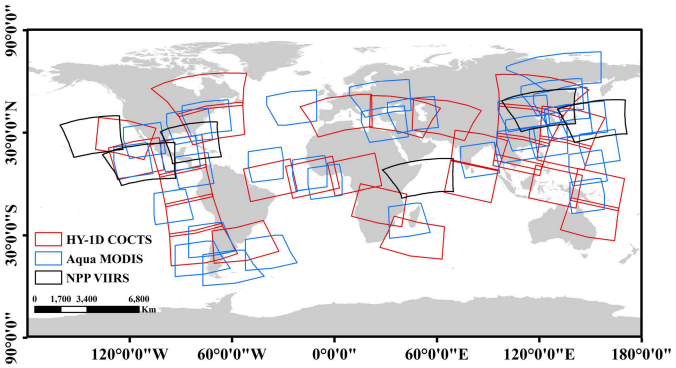


Fig. 1. Spatial distribution of satellite data.

the calibration results. The MODIS and VIIRS data employed in this study were obtained from the NASA Earthdata platform (https://search.earthdata.nasa.gov) and the NOAA CLASS platform (https://www.aev.class.noaa.gov), respectively. Specific band information is detailed in Table I.

For this study, 280 scenes of spatiotemporally quasi-synchronous (observation time difference <1 h) HY-1D COCTS Level 1A and Aqua MODIS Level 1B satellite images, acquired between 2023 and 2025, were collected. The spatial coverage encompasses China's coastal waters (e.g., the East China Sea, Yellow Sea, and South China Sea), the Northwest Pacific (Sea of Japan, Philippine Sea), the northern Indian Ocean (Bay of Bengal, Arabian Sea), and Western Pacific open oceans. These images spanned water types from turbid coastal zones to open ocean waters, enabling a comprehensive assessment of radiometric characteristics across various marine conditions. To comprehensively validate the proposed method, synchronous VIIRS data were also collected as an independent validation dataset. The spatial distribution of select satellite footprints is illustrated in Fig. 1.

B. Dataset Construction

To ensure sample quality, rigorous quality control was applied to the spatiotemporally coincident MODIS and COCTS data. Initially, pixels contaminated by sun glint (HIGLINT, MODGLINT), clouds (CLDICE), and sea ice (SEAICE) were screened out based on the MODIS Level 2 ocean color flags. Concurrently, the COCTS radiance data were filtered using internal quality flags to exclude interference from land (LAND), clouds (CLOUD), sea ice (SEAICE), stray light (STLIGHT), and high sun glint (HISUNGLINT). To minimize uncertainties from extreme viewing geometries, strict angular constraints were implemented, removing samples with solar zenith angles exceeding 80° or viewing zenith angles surpassing 50°. Following this, residual cloud shadows, glint artifacts, and anomalous pixels were manually removed through visual inspection. To mitigate the positional and spatial-resolution discrepancies between the target sensor (COCTS, 1100 m) and the reference sensor (Aqua MODIS, 1000 m), geometric registration and downsampling were performed. In particular, ENVI 5.6 software was utilized to implement geometric registration for MODIS and COCTS datasets, where the spatial registration error was strictly confined to within two pixels to ensure spatial collocation of the two datasets. Furthermore, the 1000-m MODIS imagery was downsampled to 1100-m resolution, namely the native spatial resolution of COCTS, to achieve consistent spatial resolution between the two sensors. Finally, to guarantee radiometric consistency, only sample pairs with a Pearson correlation coefficient above 0.5 were retained, thereby reducing biases stemming from environmental fluctuations.

Considering the spectral response discrepancies between the COCTS and MODIS sensors, this study adopted the spectral band adjustment factor (SBAF) method proposed by Chander et al. [28]. This approach utilizes hyperspectral observation data to simulate the equivalent reflectance of the target scene, thereby calculating the SBAF to correct radiometric biases resulting from relative spectral response mismatches. The calculation formula for SBAF is defined as (1). Once SBAF_i is calculated, it is applied to the observations of the reference sensor (MODIS) to generate reference radiance values consistent with the spectral characteristics of COCTS [see (2)]

$$SBAF_i = \frac{\bar{\rho}_{COCTS,i}}{\bar{\rho}_{MODIS,i}} = \frac{\dfrac{\int_{\lambda_1}^{\lambda_2} \rho(\lambda) S_{COCTS,i}(\lambda)\, d\lambda}{\int_{\lambda_1}^{\lambda_2} S_{COCTS,i}(\lambda)\, d\lambda}}{\dfrac{\int_{\lambda_1}^{\lambda_2} \rho(\lambda) S_{MODIS,i}(\lambda)\, d\lambda}{\int_{\lambda_1}^{\lambda_2} S_{MODIS,i}(\lambda)\, d\lambda}} \tag{1}$$

$$L_{MODIS,i}^{\text{adjusted}} = L_{MODIS,i}^{\text{original}} \times SBAF_i \tag{2}$$

where SBAF_i represents the SBAF for the ith band; $\bar{\rho}_{\mathrm{COCTS},i}$ and $\bar{\rho}_{\mathrm{MODIS},i}$ denote the equivalent TOA reflectance for COCTS and MODIS in the ith band, respectively, simulated based on the hyperspectral data. In the integral calculation, λ represents the wavelength, $[\lambda_1, \lambda_2]$ is the effective coverage range of the spectral response function; $\rho(\lambda)$ is the continuous hyperspectral curve of the target scene (PACE/OCI data are utilized in this study); and $S_{\mathrm{COCTS},i}(\lambda)$ and $S_{\mathrm{MODIS},i}(\lambda)$ correspond to the spectral response functions of the COCTS and MODIS sensors in the ith band, respectively. Based on the quality-controlled and spectrally adjusted sample data, a matched COCTS-MODIS dataset comprising 357 pairs of 32×32 pixel patches was constructed, covering diverse seasons and water environments. To meet the requirements of model training, the dataset was randomly partitioned into a training set and a validation set at a ratio of 8:2.

C. Evaluation Metrics

The proposed method was evaluated using the mean squared error (mse), mean relative error (MRE), structural similarity index measure (SSIM), and spectral angle (SA). The formulas for these metrics are as follows:

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}(Y_i - X_i)^2 \tag{3}$$

$$\mathrm{MRE} = \frac{1}{N}\sum_{i=1}^{N}\frac{|Y_i - X_i|}{|X_i|} \times 100\% \tag{4}$$

$$\mathrm{SSIM} = \frac{(2\mu_X\mu_Y + C_1)(2\sigma_{XY} + C_2)}{(\mu_X^2 + \mu_Y^2 + C_1)(\sigma_X^2 + \sigma_Y^2 + C_2)} \tag{5}$$

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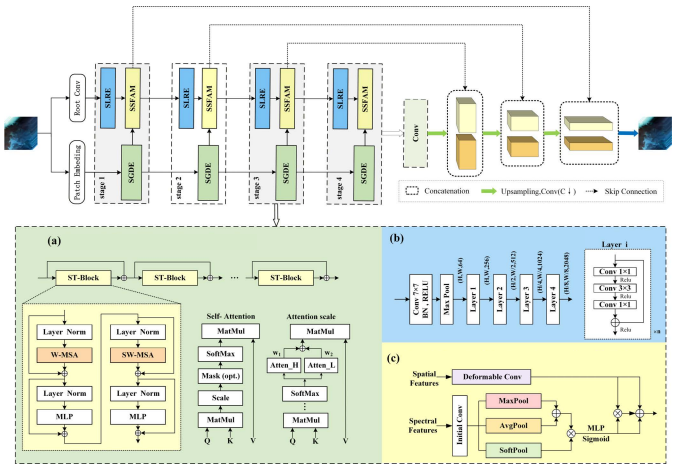


Fig. 2. Framework of DB-SSCN. (a) Spectral global dependence encoder. (b) Spatial local refinement encoder. (c) Spatial spectral feature aggregation module.

$$\mathrm{SA} = \frac{180}{\pi} \cdot \frac{1}{N} \sum_{i=1}^{N} \arccos\left(\frac{\mathbf{x}_i \cdot \mathbf{y}_i}{\|\mathbf{x}_i\| \cdot \|\mathbf{y}_i\|}\right) \tag{6}$$

where X_i and Y_i in the mse and MRE formulas denote the scalar reference and predicted values for the ith sample, respectively, with N being the total number of such pairs; μ_X and μ_Y denote the mean values of the ground truth and predictions, σ_X and σ_Y are their standard deviations, and σ_{XY} represents the covariance between them; C_1 and C_2 are stabilizing constants in the SSIM calculation. For SA, $\mathbf{x}_i$ and $\mathbf{y}_i$ are the spectral vectors (comprising all bands) of the ith spatial pixel, $\mathbf{x}_i \cdot \mathbf{y}_i$ denotes their dot product, $\|\mathbf{x}_i\|$ is the Euclidean norm, and N is the total number of spatial pixels; the SA is first computed for each pixel, and the average over all pixels yields the global mean SA, which measures the shape similarity between reference and predicted spectra by calculating the angle between spectral vectors.

III. Spatial–Spectral Collaborative Radiometric Cross-Calibration Method Based on Deep Learning

A. Overall Framework

Shifting away from the conventional paradigm of fitting linear radiometric calibration coefficients, this study designs a cross-calibration method based on a dual-branch spatial–spectral collaborative network (DB-SSCN) (Fig. 2). Tailored to the inherent 3-D spatial–spectral characteristics of satellite imagery, two parallel branches are designed to model these features. In particular, the spatial local refinement encoder (SLRE) branch leverages the inductive bias of convolutional operations. Through weight sharing and hierarchical stacking within local receptive fields, it progressively extracts multiscale spatial features, ranging from low-level fine-grained texture information to high-level semantic structural information. Conversely, the spectral global dependence encoding (SGDE) branch leverages the global context modeling capability of the self-attention mechanism, capturing long-range dependencies across all bands through query-key-value (QKV) interactions. The outputs from both branches undergo multiscale feature fusion via layerwise embedded spatial–spectral feature aggregation modules (SSFAMs) [29]. During the feature reconstruction phase, this method employs a cascaded strategy of convolution and upsampling operations instead of transposed convolutions. This design circumvents the checkerboard artifacts typically induced by kernel overlap and uneven mapping during the reconstruction process of the latter [30], thereby ensuring the spatial continuity of the calibrated results. Furthermore, addressing the issue of significant radiance range variations across different bands in multispectral imagery, a band-weighting mechanism is designed. Within the loss function, higher weights are assigned to bands with narrower value ranges, such as the near-infrared (NIR) bands. This enhances their gradient contributions during backpropagation, ultimately improving the overall accuracy and stability of the multiband joint calibration.

B. Dual-Branch Spatial–Spectral Feature Extraction Module

To simultaneously preserve local spatial details and global spectral dependencies in remote sensing imagery, this study constructs a dual-branch spatial–spectral feature extraction framework comprising SLRE and SGDE. This architecture employs two parallel pathways to extract multiscale local spatial features and long-range spectral dependencies, respectively, achieving complementary and collaborative representation of spatial–spectral characteristics. The spatial local refinement branch employs a multilayer stacked residual network structure. Through bottleneck structures composed of sequential convolutional operations (1×1, 3×3, and 1×1), it achieves efficient extraction of local spatial features while reducing computational complexity. The fundamental residual unit can be expressed as follows:

$$y_1 = \sigma\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times1}(x)\right)\right) \tag{7}$$

$$y_2 = \sigma\left(\mathrm{BN}\left(\mathrm{Conv}_{3\times3}(y_1)\right)\right) \tag{8}$$

$$y_3 = \mathrm{BN}\left(\mathrm{Conv}_{1\times1}(y_2)\right) \tag{9}$$

$$z = \sigma(y_3 + \mathcal{F}(x)) \tag{10}$$

where x represents the input features, $\mathrm{Conv}_{k\times k}$ denotes a convolutional layer with kernel size k, BN represents batch normalization, σ is the activation function, and $\mathcal{F}$ represents the 1×1 convolutional projection transformation. This branch leverages the local receptive field and weight-sharing mechanism intrinsic to convolution operations to progressively expand the effective receptive field during layerwise downsampling, effectively capturing multiscale spatial features ranging from low-level textures to high-level semantics. The residual connections mitigate the vanishing gradient problem, ensuring stable training of deep networks.

However, the intrinsic locality constraint of convolutional operations limits their capacity to capture long-range spectral dependencies. To address this, the spectral feature extraction branch introduces a self-attention-based global spectral dependence encoding mechanism to model global correlations between bands. This branch leverages an enhanced Swin Transformer architecture [see Fig. 2(a)], facilitating cross-window information interaction through a shifted window

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multihead self-attention (SW-MSA) mechanism. The window MSA (W-MSA) and SW-MSA are applied alternately in consecutive Swin Transformer blocks, establishing long-range dependencies while maintaining $O(N)$ computational complexity [31]. The detailed process is as follows:

$$\widehat{s}_l = \text{W-MSA}\left(\text{LN}\left(s_{l-1}\right)\right) + s_{l-1} \tag{11}$$

$$s_l = \text{MLP}\left(\text{LN}\left(\widehat{s}_l\right)\right) + \widehat{s}_l \tag{12}$$

$$\widehat{s}_{l+1} = \text{SW-MSA}\left(\text{LN}\left(s_l\right)\right) + s_l \tag{13}$$

$$s_{l+1} = \text{MLP}\left(\text{LN}\left(\widehat{s}_{l+1}\right)\right) + \widehat{s}_{l+1} \tag{14}$$

where s_l and s_{l+1} represent the output features of layers l and $l+1$, respectively.

To enhance feature representation capability and mitigate the issue of high-frequency information decay in deep encoding layers, an attention scale module is designed [see Fig. 2(a)]. This module decomposes the attention matrix into a low-frequency component representing smooth trends and a high-frequency component capturing local details, then dynamically adjusts their relative contributions through learnable weighting parameters to optimize feature weight distribution. The detailed mechanism is as follows:

$$A\left(Q,K,V\right) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{15}$$

$$\text{Atten}_l\left(A\right) = \frac{\mathbf{1}_{N\times N}}{N^2}\cdot\sum_{i=1}^{N}\sum_{j=1}^{N}A_{ij} \tag{16}$$

$$\text{Atten}_h\left(A\right) = A - \text{Atten}_l\left(A\right) \tag{17}$$

$$A' = w_1\cdot\text{Atten}_l\left(A\right) + w_2\cdot\text{Atten}_h\left(A\right) \tag{18}$$

where Q, K, V are derived from input spectral features through linear transformations, $A \in \mathbb{R}^{N\times N}$ represents the original attention matrix, and $\mathbf{1}_{N\times N} \in \mathbb{R}^{N\times N}$ denotes a matrix with all elements equal to 1. $\text{Atten}_l(A)$ and $\text{Atten}_h(A)$ represent the low-frequency and high-frequency components extracted from A, respectively. In particular, $\text{Atten}_l(A)$ captures smooth background information of the attention distribution by computing the global average of the attention matrix, while $\text{Atten}_h(A)$ captures local detail fluctuations relative to the average level through residual calculation. The learnable weighting coefficients $w_1, w_2 \in [0,1]$ satisfy the constraint $w_1 + w_2 = 1$, and $A' \in \mathbb{R}^{N\times N}$ denotes the reweighted output attention matrix.

C. Feature Reconstruction Module

In the feature reconstruction stage, the proposed method employs a progressive upsampling and feature transformation cascade to restore low-resolution feature maps to the original image dimensions while optimally conserving spectral and spatial structural integrity (Fig. 2). This architecture enables high-fidelity spatial–spectral reconstruction through smooth spatial frequency transitions and deep integration of multiscale features. The module discards transposed convolution operations, which are susceptible to checkerboard artifacts, opting instead for an upsampling strategy that combines bilinear interpolation with convolutional processing. In particular, the reconstruction process is defined in (19). Each reconstruction layer first applies a 3×3 convolution and ReLU activation function to the input low-resolution features to perform nonlinear transformation, thereby enhancing their representational capacity. Subsequently, the features are upsampled by a factor of 2 using bilinear interpolation to smoothly expand their spatial scale. The upsampled features are then concatenated along the channel dimension with correspondingly scaled features from the encoder via skip connections, thereby incorporating the multiscale spatial and spectral features acquired by the dual-branch encoder and enhancing the detail preservation capability of the reconstructed output. Finally, the fused features undergo another 3×3 convolution for channel alignment and integration, producing the refined high-resolution output

$$F_i^{dec} = \text{ReLU}\left(\text{Conv}_2\left(\text{Concat}\left[\uparrow_2\left(\text{Conv}_1\left(F_{i-1}^{dec}\right)\right), F_i^{enc}\right]\right)\right) \tag{19}$$

where F_i^{dec} and F_i^{enc} denote the output features of the ith decoder layer and the skip connection features from the encoder, respectively; Conv_1 and Conv_2 represent 3×3 convolutions that maintain and halve the number of channels, respectively; $\uparrow_2$ denotes the $2\times$ upsampling operation implemented via bilinear interpolation; $\text{Concat}[\cdot,\cdot]$ indicates feature concatenation along the channel dimension; and the entire reconstruction process incorporates nonlinear expressive capability through the ReLU activation function.

D. Band-Weighted Loss Function

Due to the significant disparity in radiance value ranges across different bands in ocean color satellite data, individual bands with narrower value ranges (e.g., NIR bands) suffer from insufficient gradient contribution during backpropagation. To address this issue and optimize the spectral balance during the backpropagation process, this study proposes a modified mse loss function based on band-specific weighting. By introducing band-specific weight parameters, this function effectively balances the contribution of each band during training, as defined in the following equation:

$$\mathcal{L} = \frac{1}{C}\sum_{c=1}^{C}w_c\cdot\frac{1}{H\cdot W}\sum_{i=1}^{H}\sum_{j=1}^{W}\left(\hat{y}_{c,i,j} - y_{c,i,j}\right)^2 \tag{20}$$

where $\hat{y}_{c,i,j}$ represents the model prediction, $y_{c,i,j}$ denotes the reference value, and w_c represents the predefined band-specific weighting coefficient based on radiance value range differences. C, H, and W denote the number of spectral bands, the height, and the width of the input patch, respectively. In the implementation, the band-specific weight coefficients for the eight COCTS bands are empirically set to $[1,1,1,1,1,3,4,6]$. The determination of these specific values is guided by an empirical strategy based on the inverse dynamic range of the radiometric data. Due to the physical characteristics of ocean color observation, visible bands (Bands 1–5) typically possess wider dynamic ranges and higher radiance magnitudes, whereas NIR bands (Bands 6–8) exhibit significantly narrower dynamic ranges. To prevent the gradient contributions of the NIR bands from being overwhelmed during backpropagation, the weights are initially estimated using the approximate inverse ratio of their statistical value ranges. Subsequently, these baseline values are empirically fine-tuned to the current

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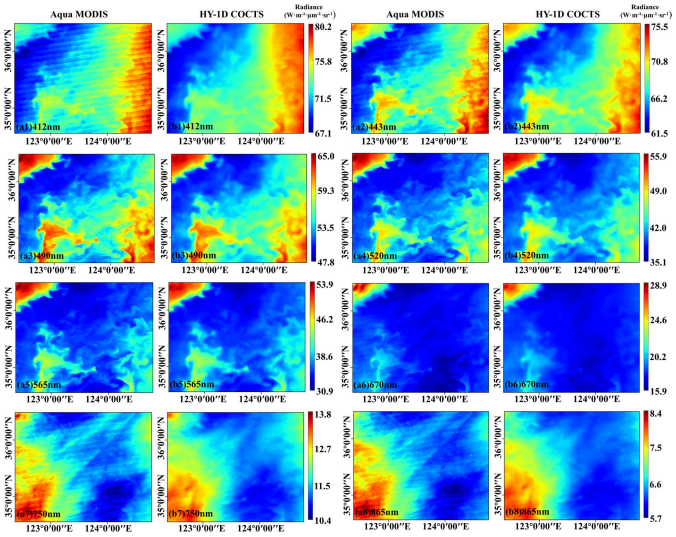


Fig. 3. Comparison between (a) MODIS radiance and (b) HY-1D COCTS radiometrically calibrated results in the Yellow Sea on March 19, 2024.

integer proportions to avoid gradient instability associated with strict exact reciprocals, thereby achieving an optimal balance between multiband gradient propagation and network convergence.

The loss function first computes the per-pixel mse across all spatial positions for each band. Subsequently, it performs a weighted summation of the loss values across the bands, ultimately obtaining the global loss value through an arithmetic average to ensure gradient stability. Through differential weighting in the spectral domain, the proposed method effectively mitigates the insufficient optimization caused by the narrower radiance ranges of the NIR bands. This approach not only ensures reconstruction accuracy in the visible bands but also significantly enhances the radiometric precision in longer-wavelength regions.

IV. Result

A. Experimental Settings

The proposed method was implemented using the PyTorch deep-learning framework and trained/evaluated on an NVIDIA RTX A6000 GPU. The optimizer employed was stochastic gradient descent with an initial learning rate of 0.01 and L2 regularization (weight decay coefficient of 1×10^{-4}). The learning rate adjustment strategy utilized the cosine annealing algorithm to dynamically optimize the training process. During the training phase, the batch size was set to 16 with a maximum of 1000 epochs. An early stopping mechanism was also incorporated, which would terminate training if the validation loss showed no improvement for 50 consecutive epochs, to mitigate overfitting.

B. Evaluation of HY-1D COCTS Cross-Calibration Results

To evaluate the performance of the developed cross-calibration method, a comparative analysis was conducted using synchronous Aqua MODIS Level 1B radiance products independent of the training samples. The results demonstrate

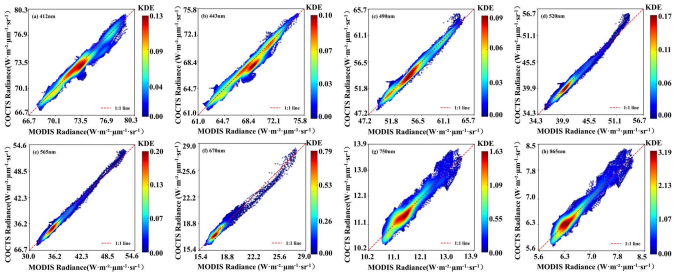


Fig. 4. Statistical results of HY-1D COCTS radiometric cross-calibration in the Yellow Sea on March 19, 2024. The colorbar represents the kernel density estimate (KDE) of the data points.

that the COCTS radiometric calibration generated by the proposed DB-SSCN method exhibits strong consistency with the MODIS radiance products. In particular, the dynamic ranges and distributions of radiance across all bands align closely, and local detailed features are effectively restored (Fig. 3). Statistical analysis of the radiance distribution reveals that the calibrated HY-1D COCTS data are highly linearly correlated with the MODIS data. The radiance distributions of the eight calibrated COCTS bands closely align with the MODIS data, with the vast majority of data points tightly clustered around the 1:1 line (Fig. 4).

Furthermore, quantitative evaluation based on the MODIS radiance products (Table II) verifies the accuracy of the COCTS calibration results, indicating an MSE of 0.2661 and an MRE of 1.18% across the eight bands. Among these, the blue bands achieve the optimal accuracy (mse: 0.448, MRE: 0.7%), whereas the NIR bands exhibit relatively lower accuracy (mse: 0.02, MRE: 1.8%). The cross-calibrated imagery likewise maintains excellent spatial structural and spectral consistency, yielding a SSIM of 0.9879 and a SA of 0.28°. These experimental results demonstrate that the proposed method achieves high-precision radiometric calibration across all eight bands of the HY-1D COCTS while effectively preserving the spatial structures and spectral shape characteristics of the imagery.

C. Comparative Analysis With VIIRS Radiance Products

To further evaluate the applicability and accuracy of the proposed calibration method, a cross-comparison analysis was conducted using SNPP VIIRS radiance products. Given the physical discrepancies in spectral band configurations between the COCTS and VIIRS sensors, the constructed SBAFs were applied to perform spectral normalization corrections on five VIIRS bands (412, 445, 488, 672, and 865 nm) to match the corresponding five COCTS bands (412, 443, 490, 670, and 865 nm). The comparison results indicate that the COCTS radiometric calibration results exhibit a consistent trend with the VIIRS observations. Although COCTS exhibits a slight systematic negative bias, the MRE across the five channels ranged from 4.47% to 10.5% after SBAF correction, with the SA of 1.35° (Fig. 5).

To investigate the origin of this systematic bias, synchronous observations from Aqua MODIS and SNPP VIIRS were further compared [see Fig. 5(b)]. The results reveal that MODIS also exhibits a similar systematic negative bias

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TABLE II
ACCURACY ASSESSMENT RESULTS BASED ON AQUA MODIS

Band(nm)	412	443	490	520	560	670	750	865
SSIM	0.9884	0.9894	0.9841	0.9821	0.9775	0.9922	0.9947	0.9951
MSE	0.4448	0.4071	0.4096	0.3605	0.3004	0.1324	0.0471	0.0230
MRE(%)	0.70	0.75	0.93	1.14	1.16	1.49	1.49	1.80
SA(°)	0.28							

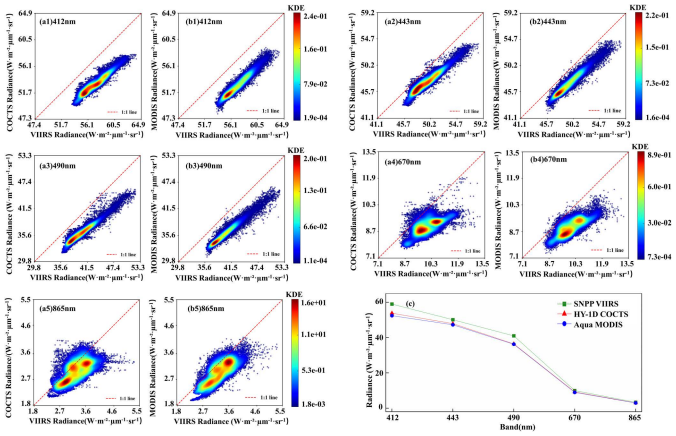


Fig. 5. Comparison of (a) HY-1D radiometric calibration results, (b) MODIS radiance products, and (c) SNPP VIIRS radiance and spectral curves. (a1) and (b1) 412 nm. (a2) and (b2) 443 nm. (a3) and (b3) 490 nm. (a4) and (b4) 670 nm. (a5) and (b5) 865 nm.

relative to VIIRS. Previous studies on water scenes have demonstrated that, influenced by factors such as sensor calibration baselines and radiative transfer models, TOA radiances measured by VIIRS are inherently elevated by approximately 7% relative to Aqua MODIS across reflective solar bands [32]. Given that the COCTS cross-calibration in this study uses Aqua MODIS as the primary reference standard, this inherent 7% baseline discrepancy between the reference and validation sensors directly propagates to and manifests in the COCTS-VIIRS comparison.

Considering this 7% systematic baseline offset, along with confounding factors under nonstrict SNO conditions—such as changes in solar-viewing geometry and dynamic variations in the marine–atmospheric environment induced by an 1-h overpass time lag—the maximum MRE of 10.5% lies well within a physically reasonable expectation and an acceptable engineering range.

Regarding the spectral shape, the calibrated COCTS mean radiance spectra remain in high agreement with those of both SNPP VIIRS and Aqua MODIS [see Fig. 5(c)].

D. Comparison With Standard COCTS Radiance Products

Furthermore, the HY-1D COCTS radiance products derived using the proposed method were compared with the official standard products. The comparison results (Fig. 6) demonstrate that, although both exhibit broadly consistent overall distribution trends, the calibration results of the proposed method more closely approximate the MODIS radiance across all spectral bands. Quantitative evaluation reveals that the proposed method achieves superior agreement with the

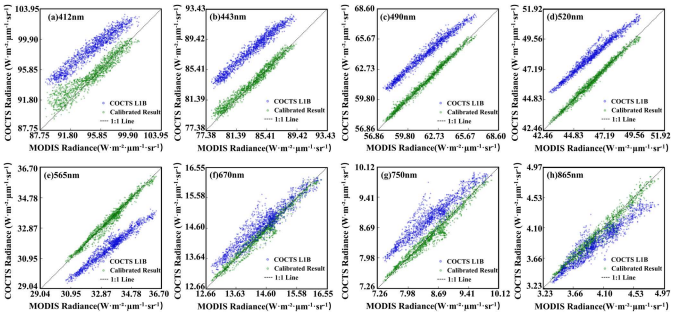


Fig. 6. Scatter plot comparison between COCTS calibrated results (green points) and standard radiance products (blue points) in the Japan Sea on September 8, 2024: (a) 412, (b) 443, (c) 490, (d) 520, (e) 565, (f) 670, (g) 750, and (h) 865 nm.

TABLE III
COMPARISON OF GLOBAL METRICS FOR DIFFERENT CALIBRATION MODELS

Calibration Method	MSE	RE (%)	SA (°)
OLS	27.40	16.15	3.74
MLP	6.22	4.60	1.88
DB-SSCN	0.30	1.62	0.30

MODIS products: the MRE across all eight bands is below 1.3%, with an average of 0.65% and an SA of 0.23°. On the contrary, the standard products exhibit larger discrepancies, with MRE values ranging from 1.99% to 5.16%, an average MRE of 4.26%, and an SA of 1.22°.

E. Comparative Analysis With Different Calibration Baseline Models

To further quantitatively evaluate the performance advantages of the proposed DB-SSCN model in cross-calibration tasks, two representative methods are introduced as baselines: the classical ordinary least squares (OLSs) linear regression model and a data-driven, pixelwise MLP model. As summarized in the global evaluation metrics in Table III and the band-by-band metrics in Tables IV and V, distinct performance variations are observed among the three models. The OLS regression yields the highest global mse, RE, and SA values (27.40, 16.15%, and 3.74°, respectively), indicating that a purely linear model struggles to fully characterize the nonlinear radiometric residuals caused by differences in sensor spectral response functions and complex ocean-atmospheric environments. Although the MLP model mitigates calibration errors to some extent by incorporating nonlinear mapping, achieving a global mse of 6.22, RE of 4.60% and SA of

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TABLE IV
COMPARISON OF MSE FOR DIFFERENT CALIBRATION MODELS

Band (nm)	412	443	490	520	560	670	750	865
OLS	36.67	32.54	33.06	16.28	51.06	26.47	16.75	6.36
MLP	10.06	5.24	16.82	3.55	12.73	0.73	0.39	0.25
DB-SSCN	0.29	0.37	0.43	0.42	0.42	0.28	0.18	0.08

TABLE V
COMPARISON OF RE FOR DIFFERENT CALIBRATION MODELS

Band (nm)	412	443	490	520	560	670	750	865
OLS	7.19	7.38	9.08	9.01	17.25	22.85	27.21	29.25
MLP	2.93	2.86	10.09	4.71	2.99	3.65	3.96	5.62
DB-SSCN	0.59	0.74	1.00	1.33	1.49	2.23	2.59	2.96

1.88°, the proposed DB-SSCN model achieves the best overall performance. In particular, its global mse and RE are reduced to 0.30 and 1.62%, respectively, while the SA, which characterizes spectral shape fidelity, reaches a low of 0.30°.

V. DISCUSSION

A. Analysis of Spatial–Spectral Collaborative Modeling Mechanism

Conventional radiometric calibration methods typically employ linear calibration coefficients for each band, treating individual pixels as independent units during calibration. This approach fails to model spatial contextual relationships, thereby limiting the utilization of neighboring pixel information to suppress the interference of local anomalous pixels. Consequently, anomalous features like striping artifacts are frequently preserved or even amplified during calibration [33], [34], [35]. In the spectral dimension, conventional methods employ a band-by-band calibration strategy, neglecting interband dependencies and failing to impose constraints on the spectral shape. On the contrary, the proposed DB-SSCN method simultaneously preserves spatial structural integrity and spectral shape through a spatial–spectral collaborative modeling mechanism. In the spatial dimension, this mechanism leverages multilayer convolutions to progressively extract spatial details and construct multiscale, hierarchical feature representations. Meanwhile, the spectral branch utilizes a self-attention mechanism to establish global dependencies among bands, and incorporates a learnable attention scale module to dynamically optimize the weight distribution. This design effectively maintains critical spectral details while capturing global contextual information.

The calibration results demonstrate that the DB-SSCN method, based on the spatial–spectral collaborative mechanism, exhibits significant advantages in radiometric calibration. Through the collaborative modeling of spatial–spectral context, it enhances the consistency of the calibration results in both the spatial and spectral dimensions. In the spatial dimension, DB-SSCN effectively suppresses the interference of local anomalous pixels, such as striping and random noise, thereby improving the robustness of the calibration results (Fig. 7). An examination of the spatial profiles reveals noticeable striping artifacts in the original uncalibrated COCTS DN imagery [see Fig. 8(a)]. These striping artifacts are retained in the COCTS Level 1B standard radiance products generated by conventional linear radiometric calibration models [see Fig. 8(b)]. On the contrary, after applying the DB-SSCN calibration, the striping artifacts are effectively suppressed, and the profile data become significantly smoother [see Fig. 8(c)].

In the spectral dimension, DB-SSCN demonstrates superior spectral preservation capability. In the RGB composite images,

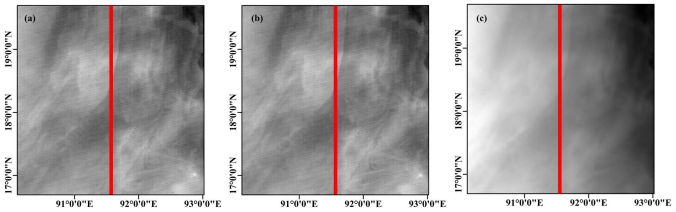


Fig. 7. Grayscale images of the 412-nm band after min–max linear stretching. (a) Uncalibrated DN image. (b) Level 1B radiance image. (c) Calibrated radiance image.

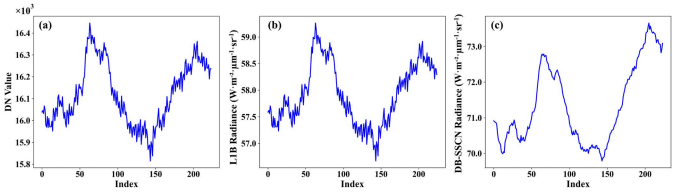


Fig. 8. Profile comparison at 412 nm among (a) uncalibrated DN image, (b) Level 1B radiance image, and (c) calibrated radiance image. The profile corresponds to the red line in Fig. 7.

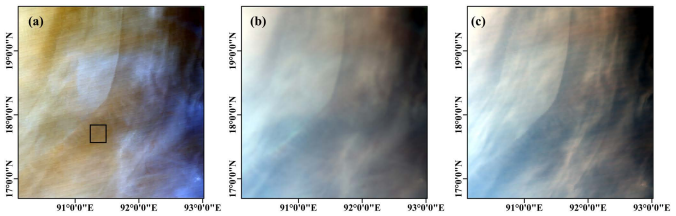


Fig. 9. RGB color composite images of (a) Level 1B radiance image, (b) calibrated radiance image, and (c) MODIS radiance image. The composite uses 670 nm (red), 490 nm (green), and 412 nm (blue).

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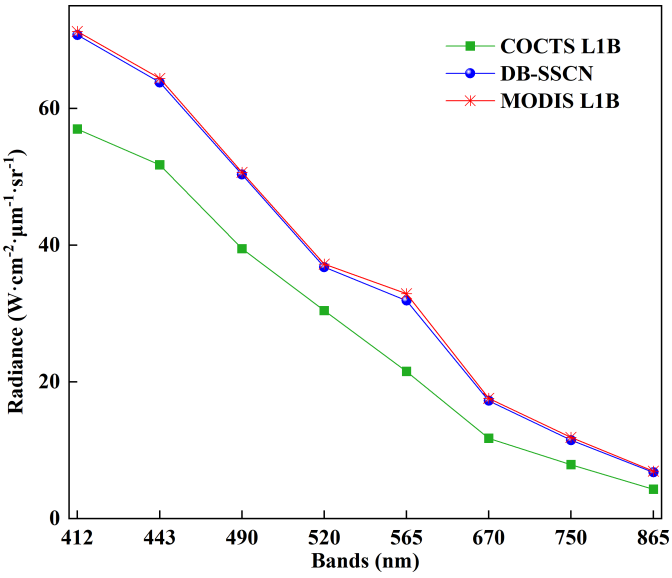


Fig. 10. Spectral curve comparison within the black box region in Fig. 9(a) for the COCTS Level 1B radiance image, the DB-SSCN calibrated radiance image, and the MODIS Level 1B radiance image.

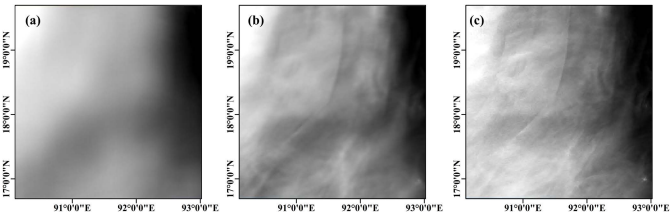


Fig. 11. Comparison of NIR band (825 nm) calibration results. (a) Unweighted loss function. (b) Weighted band-weighted loss function. (c) MODIS reference image.

the calibration results derived from DB-SSCN [see Fig. 9(b)] align more closely with the MODIS reference imagery [see Fig. 9(c)] compared to the COCTS Level 1B standard products [see Fig. 9(a)]. The region of interest marked by the black box in Fig. 9(a) was selected, and its mean spectral curve was extracted and plotted (Fig. 10). The results reveal a significant deviation between the spectral curve of the Level 1B product and the MODIS reference curve, whereas the spectrum of the DB-SSCN calibration data exhibits much better agreement with the MODIS spectral shape.

Furthermore, the band-weighting strategy proposed in this study plays a crucial role in balancing the contributions across different bands. By assigning higher weight coefficients to bands with narrower value ranges (e.g., the NIR bands), the contribution of their loss terms to the total loss is increased. Consequently, these bands receive stronger gradients during backpropagation, ensuring that relatively balanced attention is allocated to all bands throughout the training process. The calibration results demonstrate that when using the conventional mse loss, the calibrated results for the NIR band (825 nm) exhibit severe spatial blurring and a substantial loss of spatial information [see Fig. 11(a)]. On the contrary, the introduction of the band-weighting mechanism substantially improves the calibration quality for this band [see Fig. 11(b)], yielding

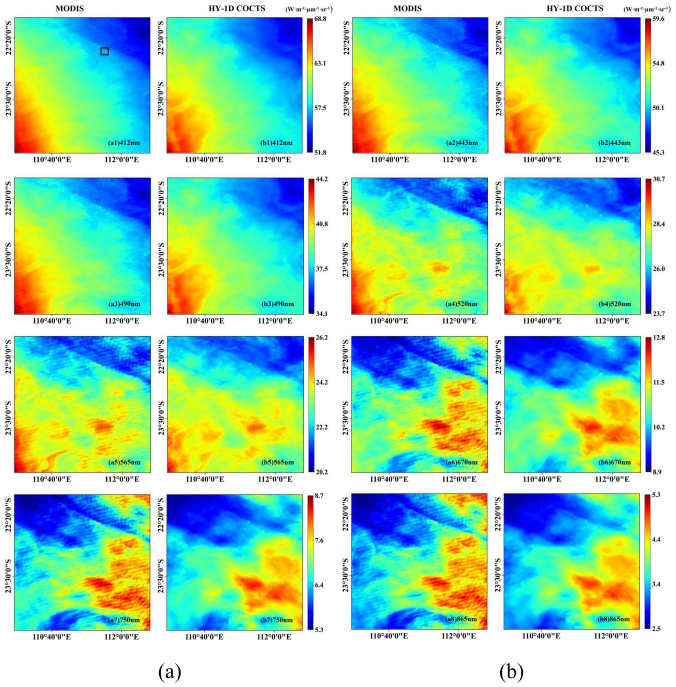


Fig. 12. Radiance comparison in clear waters of the Pacific (September 16, 2024): (a) Aqua MODIS versus (b) calibrated COCTS.

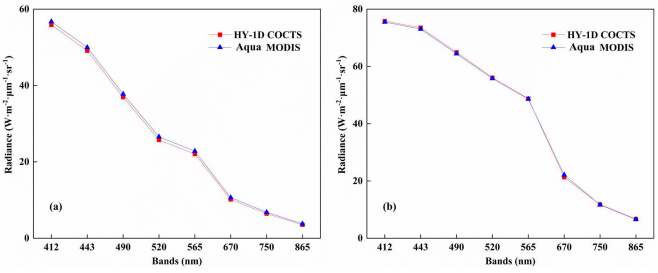


Fig. 13. Spectral curve comparison between Aqua MODIS and HY-1D COCTS under different water types. (a) Clear water samples from the Pacific. (b) Turbid water samples from the Yangtze River estuary.

richer image details and significantly enhancing the spatial consistency with the reference imagery.

B. Applicability Across Different Water Types

To evaluate the applicability of the proposed method across different water types, this study selected two typical regions for assessment: the open ocean waters of the Pacific (clear waters) and the turbid coastal waters of the Yangtze River estuary. In the clear waters of the open Pacific, the calibrated COCTS radiance exhibits strong consistency with MODIS in both spatial distribution and dynamic range (Fig. 12). Quantitative results reveal an MRE of 1.49% across the eight bands, an SSIM of 0.9954, and an SA of 0.26°, indicating high calibration accuracy. Furthermore, a comparison of the mean spectral curves from the region enclosed by the black box reveals a close agreement in spectral shape between the proposed method and MODIS [see Fig. 13(a)].

In the turbid water region, the calibrated COCTS radiance also maintains strong consistency with MODIS in terms

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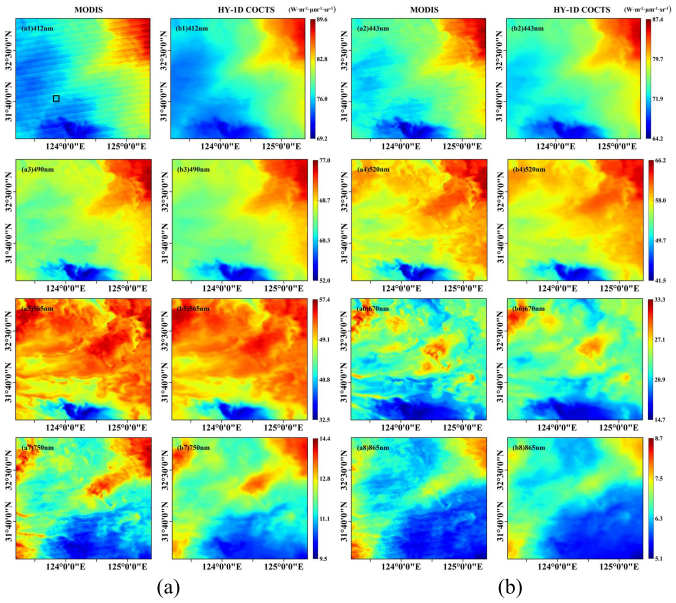


Fig. 14. Radiance comparison in turbid waters of the Yangtze River estuary (May 12, 2024): (a) Aqua MODIS versus (b) calibrated COCTS.

of spatial distribution and value range (Fig. 14). It yields an MRE of 1.33% across the eight bands, an SSIM of 0.9692, and an SA of 0.38°, confirming the method's accuracy in turbid coastal waters. The spectral curve comparison further demonstrates the effective preservation of spectral profiles under optically complex turbid water conditions Fig. 13(b).

These experimental results indicate that the proposed radiometric calibration method exhibits robust applicability and stability across different water types (including clear and turbid waters), and can provide high-accuracy radiance products for quantitative remote sensing applications in diverse water bodies.

C. Uncertainty Analysis

As a data-driven deep-learning model, DB-SSCN exhibits a cross-radiometric calibration uncertainty that is highly contingent upon the quality of the sample data. Furthermore, this uncertainty is also attributable to spectral band discrepancies, spatiotemporal matching errors, and observation geometry effects. To ensure the reliability of the calibration results, systematic analysis and mitigation were conducted in three dimensions: the data foundation, the matching process, and the physical mechanisms.

First, the quality of the reference data and the representativeness of the samples are the prerequisites that determine calibration uncertainty. On the one hand, MODIS, which possesses high-precision radiometric calibration capabilities, was selected as the reference benchmark. Its L1B product has an inherent calibration uncertainty of approximately 3% [36], making it an ideal benchmark source for cross-calibration. To reduce input noise, rigorous quality control was implemented during the preprocessing stage to exclude anomalous pixels. On the other hand, the established dataset widely covers diverse scenarios ranging from turbid coastal waters to clear oceanic waters, fully characterizing the differences in radiometric features under various marine environmental conditions. This ensures the model's generalization capability and stability across different dynamic ranges. Furthermore, addressing the discrepancies in spectral response functions between the COCTS and MODIS sensors, the SBAF was calculated to compensate for the radiometric bias introduced by spectral mismatch, further reducing the uncertainty in the data source dimension.

Second, the spatiotemporal consistency of the image pairs is a critical process variable affecting calibration accuracy. In the temporal dimension, the overpass time difference was strictly constrained to within 1 h. It is assumed that the atmospheric and water environments remain stable within this short window, thereby treating the radiometric error introduced by the time difference as negligible. In the spatial dimension, all image pairs underwent fine geometric registration, with the error constrained to within two pixels. Notably, compared to the high sensitivity of traditional point-to-point calculation methods to registration errors, the proposed DB-SSCN model is not limited to the simple mapping of single pixels. Instead, it learns the high-dimensional feature representation of images through the deep network. This feature-based calibration paradigm is capable of capturing spatial structures and contextual information within local neighborhoods, focusing on the consistency of target features rather than absolute grid alignment. This mechanism enables the model to exhibit stronger robustness against minor geometric registration deviations, thereby significantly reducing the uncertainty introduced by spatial mismatches at the algorithmic level.

Finally, observation geometry discrepancies are important physical factors leading to inconsistencies in radiometric response. Large-angle observations increase the optical path and cause significant bidirectional reflectance distribution function effects, while low solar elevation angles may introduce STLIGHT interference. To address this, strict angular constraint strategies were implemented. By excluding large-angle observation samples, instrument STLIGHT contamination and non-Lambertian effects in marginal imaging regions were effectively avoided. Through the aforementioned multidimensional error control and uncertainty mitigation strategies, the reliability and stability of the cross-calibration results were effectively ensured.

D. Application

To further evaluate the application potential of the developed method, a case study was conducted in the Yellow River estuary. As China's second-longest river and one of the world's most sediment-laden rivers, the Yellow River discharges substantial sediments into the Bohai Sea annually. This leads to significantly elevated concentrations of total suspended matter (TSM) in the estuarine and coastal waters, profoundly impacting coastal ecology, waterway stability, and biogeochemical processes [37], [38]. First, the proposed DB-SSCN method was employed to perform radiometric calibration on the HY-1D COCTS data of the study area. The calibrated radiance products were then atmospherically corrected using the composite atmospheric correction method developed by Ye et al.

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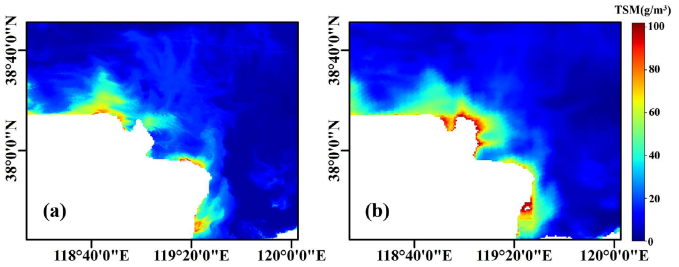


Fig. 15. Comparison of TSM concentrations in the Yellow River Estuary derived from (a) Sentinel-3 OLCI and (b) COCTS/HY-1D on February 24, 2025.

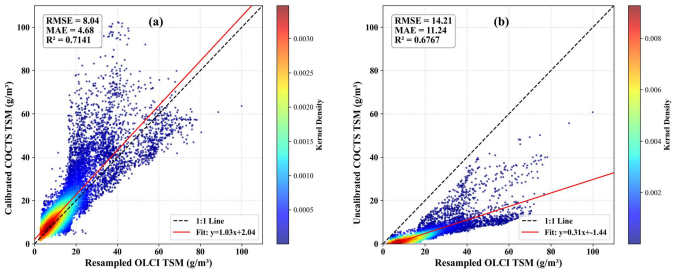


Fig. 16. Scatter comparison of Sentinel-3 OLCI-derived TSM and HY-1D COCTS-retrieved TSM in the Yellow River Estuary. (a) TSM retrieval results from calibrated COCTS. (b) TSM retrieval results from the standard COCTS product.

[39]. Based on the remote sensing reflectance derived from the atmospheric correction, a regional TSM retrieval model for the HY-1D COCTS sensor was developed by referencing the model of Doxaran et al. [40] and incorporating documented sensitive bands [41], [42].

Subsequently, the TSM concentration in the Yellow River estuary was retrieved and compared with the Sentinel-3 Ocean and Land Color Instrument (OLCI) TSM products (Fig. 15). Given the substantial difference in spatial resolution between Sentinel-3 OLCI (300 m) and HY-1D COCTS (1100 m), a spatial mean resampling strategy was adopted to register the OLCI data onto the same spatial grid as COCTS, enabling a scale-consistent comparison. To ensure the objectivity of the comparison, identical atmospheric correction and TSM inversion algorithms were applied to both datasets. Statistical results show that the calibrated COCTS-derived TSM results achieved an RMSE of 8.04 g/m^3, an R^2 of 0.71, and an MAE of 4.68 g/m^3; whereas the uncalibrated standard product yielded an RMSE of 14.21 g/m^3, an R^2 of 0.68, and an MAE of 11.24 g/m^3. These results indicate that cross-radiometric calibration significantly improves the TSM retrieval accuracy of COCTS. Furthermore, the density scatter plots (Fig. 16) demonstrate that, compared with the standard calibrated product, the corrected COCTS data exhibit a more concentrated distribution around the 1:1 reference line, indicating that radiometric calibration effectively reduces intersensor systematic biases and improves the consistency between COCTS-derived TSM retrievals and OLCI reference products.

However, some residual discrepancies remain after calibration due to differences in sensor observation conditions and retrieval processes. In particular, Sentinel-3 OLCI and HY-1D COCTS are onboard morning and afternoon satellites, respectively, resulting in unavoidable differences in overpass time. Meanwhile, the Yellow River estuary is a typical high-turbidity water region, where TSM concentrations are strongly influenced by tidal variations, hydrodynamic processes, and short-term suspended matter transport. In addition, spatial resampling uncertainties caused by different sensor spatial resolutions and model discrepancies between independent inversion algorithms may further contribute to the comparison errors. Overall, the quantitative metrics and spatial distribution characteristics demonstrate that radiometric calibration can substantially improve the practical accuracy of COCTS data.

VI. Conclusion

This study proposes a dual-branch spatial–spectral collaborative cross-calibration method. Departing from conventional linear-coefficient-based modeling, this approach deeply mines high-dimensional image features to explore a novel, explicit-coefficient-free paradigm for cross-calibration, directly realizing the conversion from DN values to TOA radiance. The method constructs a dual-branch architecture by incorporating convolution and self-attention mechanisms to extract multiscale spatial features and long-range spectral dependencies, respectively, achieving complementarity and synergy of spatial and spectral information through a feature fusion module. Moreover, a band-weighted loss function is designed to address the significant variations in radiance value ranges across different bands, which substantially improves the calibration performance for the NIR bands. Validation against synchronous Aqua MODIS radiance products demonstrates that the DB-SSCN method exhibits exceptional calibration performance. Furthermore, it effectively preserves the spatial structure and spectral shapes of the imagery during calibration, successfully suppressing the interference of local anomalies and striping artifacts. Evaluations in diverse water environments confirm its robustness and broad applicability to both turbid coastal and clear open ocean waters.

However, the proposed method has certain limitations. As a data-driven model, it is currently applicable only to HY-1D COCTS. Subsequent application to other satellites would require transfer learning of the model.

Acknowledgment

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